Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number - 13

Start date – 02/02/2026

Names: A. Vedavarshini

Anwesha Barpanda

V. Sathvik kumar

Nivetha

Kashish Patel

Marasu Vijaya Lakshmi

Internship Duration: 8 Weeks

1. Project Title

HotelRevAI — AI-Driven Revenue Analysis for Hotels

2. Project Objective

The HotelRevAI project aims to leverage data analytics and Power BI to transform raw hotel booking data into actionable business intelligence. Specifically, the project was designed to:

- Analyse overall revenue performance and identify growth trends over time.
- Evaluate hotel KPIs including ADR, cancellation rate, total bookings, and guest counts.
- Analyse customer behaviour and booking patterns by customer type and distribution channel.
- Develop DAX measures and a structured Business Matrix framework for all analytical dimensions.
- Create four interactive Power BI dashboards with dynamic filtering by year, hotel type, and customer type.
- Enable data-driven decision-making in pricing, occupancy management, and customer loyalty strategy.
- Quantify the financial impact of cancellations and identify revenue recovery opportunities.
- Support hotel management with seasonal demand analysis and stay pattern insights.

3. Project description in detail

HotelRevAI is a Power BI-based analytics project designed to address the hospitality industry's core challenge: extracting meaningful intelligence from high-volume transactional booking data. Hotels generate large quantities of daily booking records from multiple channels including direct reservations, online travel agencies (OTAs), and corporate accounts. Without structured visualization and analytical frameworks, it is difficult for management to identify revenue opportunities, understand customer behaviour, or respond to seasonal demand shifts.

The project uses a hotel booking dataset sourced from Kaggle, covering bookings for both a City Hotel and a Resort Hotel. The dataset contains over 75,000 booking records spanning multiple years (2015–2017), including fields such as arrival date,

length of stay, number of guests, distribution channel, customer type, Average Daily Rate (ADR), and cancellation status.

Approach & Methodology

The project followed a structured analytics pipeline:

- Data Ingestion & Preprocessing: The CSV dataset was imported into Power BI. Power Query Editor was used to handle missing values, correct data types, remove duplicates, and create derived columns including Total Stay Duration, Total Guests, Revenue (ADR × Total Nights), and a consolidated Arrival Date field.
- Data Modelling: A star schema was designed with Hotel_Data as the central fact table, a Calendar dimension for time-based intelligence, and a dedicated d3_measures table for all computed DAX metrics. A one-to-many relationship was established between Calendar[Date] and Hotel_data[Arrival Date].
- DAX Measure Library: Sixteen DAX measures were developed covering revenue, bookings, cancellations, customer behaviour, and occupancy. These include Total Revenue, Cancellation Rate, Revenue Loss, Repeat %, Occupancy Proxy, Revenue per Night, and more.
- Business Matrix Framework: Every DAX measure was mapped to a strategic business dimension — Financial Performance, Customer Behaviour, Occupancy & Demand, Booking Patterns, and Cancellation Management — ensuring each metric serves a defined decision-making purpose.
- Dashboard Development: Four interactive Power BI dashboards were built: Hotel Revenue Performance, Customer and Booking Behaviour, Revenue Optimization and Key Insights, and Seasonality and Stay Pattern Analysis.

Technology Used

- Microsoft Power BI Desktop — primary analytics and visualization platform
- Power Query Editor — data cleaning, transformation, and preprocessing
- DAX (Data Analysis Expressions) — custom measure development
- Kaggle Hotel Bookings Dataset — real-world transactional data
- Star Schema Data Modelling — relational data architecture

Real-World Impact

HotelRevAI demonstrates how data visualization and business intelligence tools can transform raw booking records into a strategic asset. The dashboards empower hotel managers to optimize pricing during peak seasons, reduce revenue loss from cancellations, identify high-value customer segments, and develop targeted marketing campaigns. The project directly illustrates how data-driven decision-making can improve efficiency and revenue performance.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understanding project topic and problem statement	Studied hotel revenue analysis concept and identified the project scope.
Week 2	Dataset collection and understanding columns	Collected hotel booking dataset and analyzed important fields such as hotel type, ADR, cancellation status, and arrival date.
Week 3	Data cleaning and preprocessing	Cleaned the dataset using Power Query, corrected formats, and removed missing/duplicate values.
Week 4	Data modeling design	Created star schema model with fact and dimension tables.
Week 5	DAX calculations and KPI creation	Created measures like Total Revenue, Total Bookings, and Average Stay.
Week 6	Dashboard design	Developed visual reports
Week 7	Insight generation and data storytelling	Interpreted the data, identified trends, and prepared meaningful insights.
Week 8	Final review and presentation preparation	Finalized the dashboard, reviewed project

		outcomes, and prepared presentation.

5a. Key Milestones

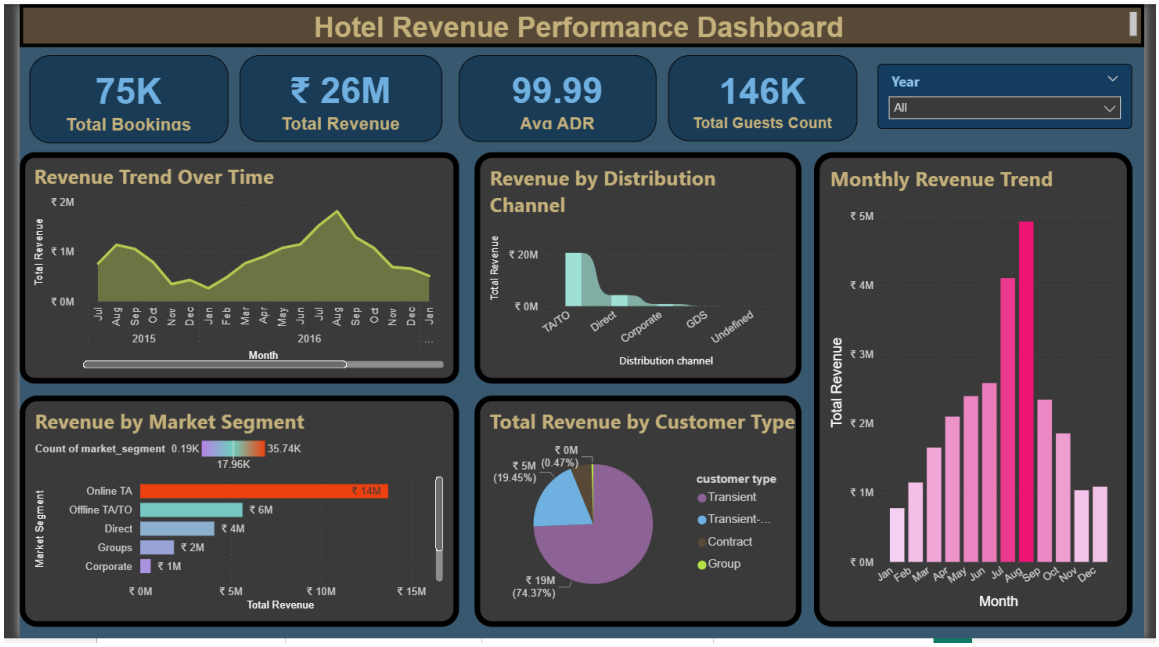
Milestone	Description	Date Achieved
Project Kickoff	Started internship project and understood the problem statement	5-02-2026
Prototype/First Draft	Completed initial data cleaning and created basic dashboard draft	11-02-2026
Mid-Term Review	Reviewed progress in data modeling, DAX, and KPI implementation	20-02-2026
Final Submission	Completed dashboard and report documentation	13-03-2026
Presentation	Presented HotelRevAI project findings and outcomes	24-03-2026

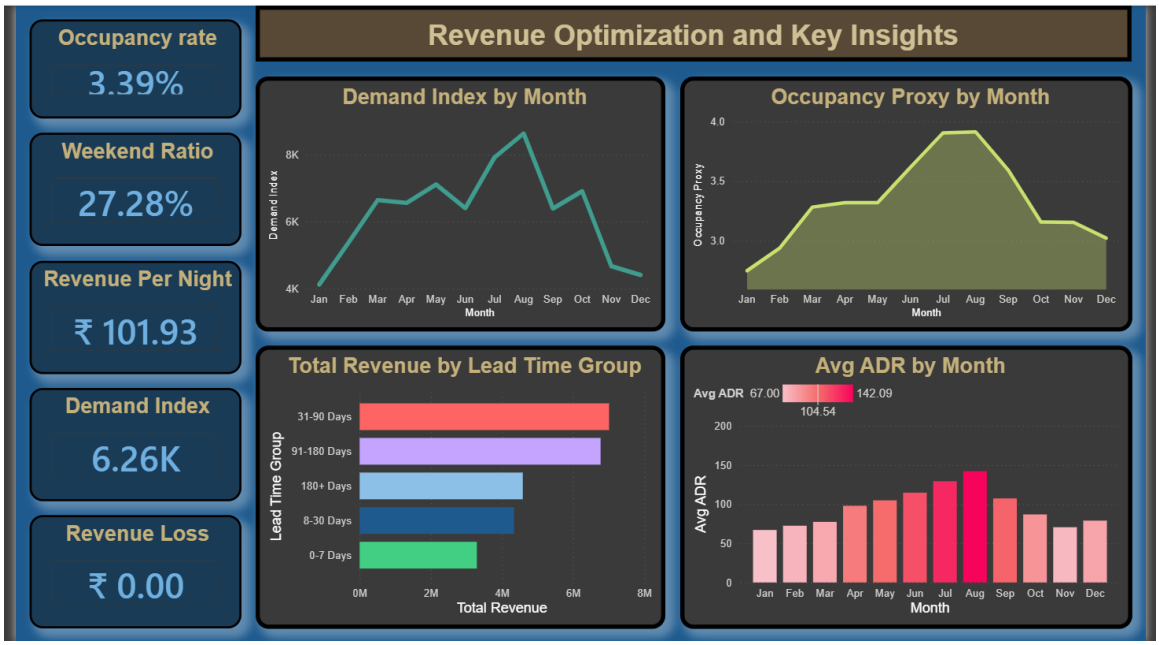
5b. Project execution details

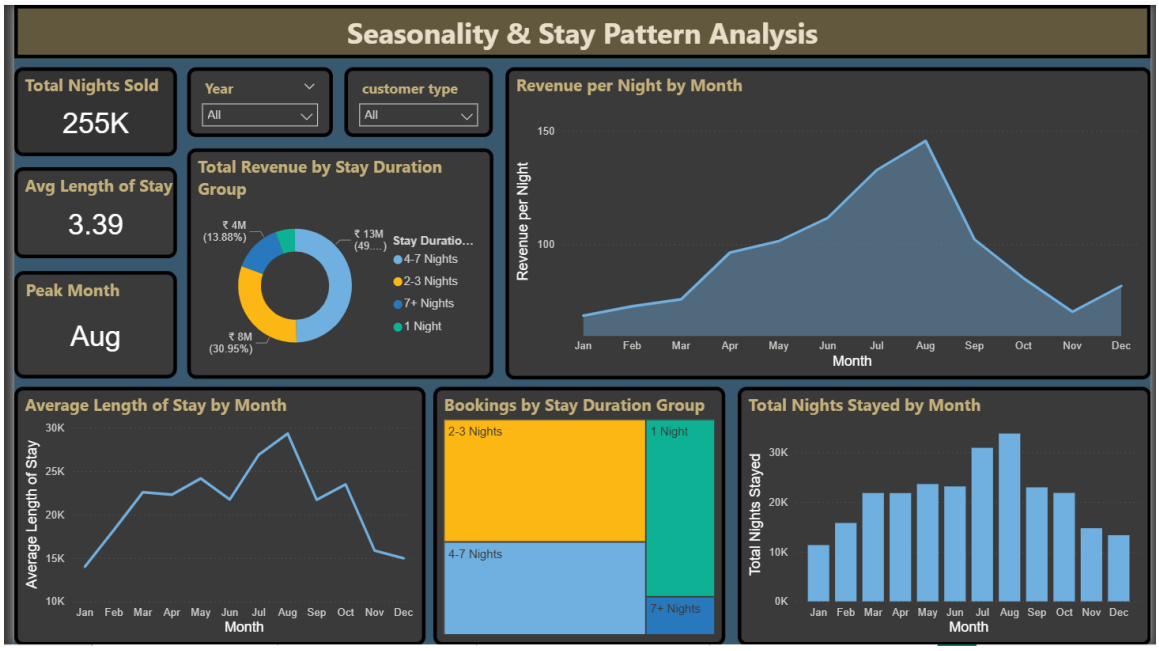
The HotelRevAI project was carried out over an 8-week internship in structured phases. Initially, the team focused on understanding hospitality revenue challenges and sourcing a relevant Kaggle dataset, identifying key business-related fields. This was followed by data preprocessing in Power BI, where cleaning, transformation, and creation of derived columns like revenue and total stay were performed. In the modeling phase, a star schema was designed with a fact table and a calendar dimension to enable time-based analysis. Next, 16 DAX measures were developed to support dynamic and accurate KPI calculations within a defined business matrix. The team then created four interactive dashboards focused on revenue performance, customer behavior, optimization,

and seasonality insights. Finally, the project was reviewed, documented, and presented with key findings and business recommendations.

6. Snapshots / Screenshots







7. Challenges Faced

The HotelRevAI project presented several technical, analytical, and design challenges throughout its execution:

- Data Quality and Missing Values: The dataset contained null or missing values in critical fields such as country, agent, and company. Identifying the

correct handling strategy — whether to fill, exclude, or transform — required careful judgment to avoid distorting analytical results.

- Data Type Corrections and Date Consolidation: The arrival date was stored across three separate columns (year, month, day). Consolidating these into a single Arrival Date column required custom transformation steps in Power Query, along with careful data type standardization.
- Star Schema Design: Designing an efficient relational data model required understanding the distinction between fact and dimension tables and establishing correct cardinality. Ensuring the Calendar dimension relationship functioned correctly across all four dashboards was a key technical challenge.
- DAX Measure Complexity: Writing accurate DAX measures — particularly those involving CALCULATE, DIVIDE, and FILTER functions — required a thorough understanding of Row Context and Filter Context. Measures such as Revenue Loss and Cancellation Impact needed careful validation against the underlying data.
- Dashboard Design and Visual Clarity: Presenting complex multi-dimensional insights in an intuitive, visually consistent format required significant iteration. Balancing information density with usability, and ensuring slicers filtered correctly across all visuals, demanded repeated testing.
- Business Story Extraction: Translating quantitative metrics into meaningful business narratives required domain knowledge of hotel revenue management, including concepts like ADR, RevPAR, occupancy optimization, and lead time pricing strategy.

These challenges were addressed through iterative testing, peer review within the team, careful data validation, and continuous refinement of both the data model and dashboard designs.

8. Learnings & Skills Acquired

The HotelRevAI internship provided a comprehensive combination of technical, analytical, and professional skills:

Technical Skills

- Power BI Desktop — building interactive dashboards, configuring slicers, and creating multi-page reports
- Power Query Editor — data cleaning, transformation, column profiling, and derived column creation
- DAX (Data Analysis Expressions) — developing business measures using CALCULATE, DIVIDE, FILTER, SUM, COUNTROWS, and time intelligence functions

- Star Schema Data Modelling — designing fact and dimension tables, establishing model relationships, and understanding cardinality
- Business Matrix Framework — mapping KPIs to strategic business decisions

Analytical & Domain Skills

- Hotel revenue management concepts: ADR, RevPAR, occupancy rate, cancellation impact, and lead time analysis
- Customer segmentation analysis: distinguishing transient, group, contract, and transient-party booking types
- Seasonality and demand pattern analysis across multi-year datasets
- Translating raw data insights into business recommendations and strategic narratives

Professional Skills

- Structured project execution across a multi-week timeline with clear milestones
- Technical documentation and report writing for business and academic audiences
- Data storytelling — presenting complex analytical findings in an accessible, visually compelling format
- Problem-solving through iterative testing, validation, and dashboard refinement

9. Testimonials from team

Working on HotelRevAI was a genuinely enriching experience. The project gave me the opportunity to apply Power BI and data analytics in a real-world business scenario — moving from raw CSV data to four interactive dashboards that tell a coherent revenue story for hotel management.

The most valuable aspect of this internship was learning how to build a structured analytics pipeline from the ground up: cleaning and transforming data with Power Query, designing a proper star schema data model, writing 16 DAX measures that operate correctly across filter contexts, and then presenting the findings in dashboards that non-technical stakeholders can actually use and explore.

I also developed a much deeper appreciation for the business side of analytics. Understanding concepts like ADR, occupancy optimization, cancellation impact, and lead time pricing — and then seeing those concepts reflected directly in the dashboards — made the project feel genuinely impactful rather than just an academic exercise.

This internship significantly strengthened my confidence in business intelligence tools and my ability to deliver data-driven insights that can guide real strategic decisions. It has reinforced my interest in pursuing a career at the intersection of data analytics and business strategy.

10. Conclusion

The HotelRevAI project successfully demonstrates the application of Power BI as an end-to-end data analytics platform for hotel revenue intelligence. Beginning with structured data loading and preprocessing, through relational data modelling and a comprehensive DAX measure library, to the development of four interactive dashboards — the project delivers a complete analytical solution that transforms raw booking data into actionable business insights.

The four dashboards collectively tell a coherent business story: Online Travel Agency channels and Transient customers drive the majority of revenue; August is the consistent peak season and the most critical period for dynamic pricing; the repeat guest rate (4.33%) represents an untapped loyalty opportunity; and 4–7 night stays contribute disproportionately to revenue despite shorter stays dominating booking counts.

Key findings from the analysis include: total revenue of ₹26M across 75,000 bookings; August as the peak revenue month approaching ₹5M; Online TA/TO as the dominant channel at ₹20M+; a very low repeat guest rate signalling a significant retention opportunity; and a clear seasonal revenue curve rising from ₹65/night in January to ₹145/night in August.

The Business Matrix framework established in this project ensures that every analytical metric is tied to a concrete strategic decision — whether that is optimizing pricing by season, improving cancellation policies, developing loyalty programmes, or expanding direct booking channels. Together, these deliverables illustrate how data-driven intelligence can meaningfully improve operational efficiency and revenue performance in the hospitality industry.

11. Acknowledgements

We would like to express our sincere gratitude to **Infosys Springboard** for providing us with this valuable internship opportunity. This internship helped us gain practical exposure to data analytics, Power BI, and business intelligence concepts in a real-world problem domain.

We would also like to thank our mentors, guide **Selciya Mam** and team members for their continuous support, encouragement, and valuable suggestions throughout the project. Their guidance helped us successfully complete the HotelRevAI project and gain confidence in applying data-driven approaches to business problems.